

(c) The class `Tree` stores the tree.

Tree	
<code>FirstNode : Node</code>	stores the root node in the tree
<code>Constructor()</code>	initialises <code>FirstNode</code> to its parameter value
<code>GetRootNode()</code>	returns the node stored in <code>FirstNode</code>
<code>Insert()</code>	stores its parameter node in the correct position in the tree

(i) Write program code to declare the class `Tree` and its constructor.

Do **not** declare the other methods.

Use your programming language appropriate constructor.

If you are writing in Python, include attribute declarations using comments.

Save your program.

Copy and paste the program code into part **3(c)(i)** in the evidence document.

[2]

(ii) Write program code for `GetRootNode()`

Save your program.

Copy and paste the program code into part **3(c)(ii)** in the evidence document.

[1]

(iii) The method `Insert()` takes a node as a parameter and then searches the tree to find the position to insert the new node by:

- checking whether the node's data is less than or greater than the root node's data
- moving to the left node if the data is less than the root node's data
- moving to the right node if the data is greater than or equal to the root node's data
- repeating until the final position of the node is found and stores the node in that position.

Write program code for `Insert()`

Save your program.

Copy and paste the program code into part **3(c)(iii)** in the evidence document.

[6]